

Rohan Alias

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PROFILE

Results-driven Azure Data Engineer with 3.7 years of experience delivering cloud-based data solutions across the full data lifecycle. Skilled in designing and optimizing ETL/ELT pipelines, integrating diverse data sources, and building scalable solutions using Databricks, PySpark, and SQL. Proven track record of enhancing data quality, streamlining processes, and enabling data-driven decision making through reliable, efficient, and modern data engineering practices

TECHNICAL SKILLS

- **Programming Languages:** Python, Linux
- **Big Data Technologies:** PySpark, Apache Spark SQL, Databricks, Delta Lake, Delta live tables, Unity Catalog, Data Modeling, ADF, ADLS, Azure Databricks, Azure SQL Database, S3, Lambda, Athena, Glue
- **Databases & Connectors:** MySQL, Cosmos DB
- **Dashboarding & Monitoring Tools:** Power BI
- **Workflow Management:** Airflow, Control-M
- **Build Tools & Containers:** Maven, Jenkins, Docker
- **Generative AI & Advanced Analytics:** (RAG), Vector Embeddings, Azure OpenAI, Semantic Search, LLM Integration

PROFESSIONAL EXPERIENCE

Consultant (Data Engineer II)

01/2023 – present | Bangalore

Deloitte

- Designed end-to-end **ETL pipelines** using **Medallion Architecture** (Bronze, Silver, Gold layers) on **ADLS Gen2**, ingesting and processing **millions** of operational records daily to support scalable analytics.
- Engineered batch and near real-time streaming data pipelines with **Azure Databricks**, **PySpark**, and **Kafka**, enabling low-latency analytics and delivering actionable insights to stakeholders.
- Built a comprehensive data cleansing, validation, and reconciliation framework that ensured **high data quality** and standardization across pipeline layers, supported by robust logging, monitoring, and alerting.
- Leveraged **Delta Lake** for reliable, high-performance storage; applied advanced optimizations including partitioning, bucketing, salting, vacuuming, and file compaction, **reducing processing times by 30%** and lowering compute costs.
- Implemented **Unity Catalog** to enforce standardized access patterns, **row- and column-level security**, and **governance**, achieving **100% compliance** in data validation and reconciliation while improving discoverability of analytics assets across teams.
- Developed and maintained **CI/CD pipelines** to automate the deployment of data infrastructure and pipelines across environments, increasing release reliability and significantly reducing manual effort and deployment risks.
- Designed and developed interactive **Power BI** dashboards that transformed raw data into actionable business insights, streamlining stakeholder reporting and decision-making.
- Developed a **Python**-based security library using **FastAPI** to standardize authentication and authorization mechanisms, ensuring consistent security practices across multiple applications

EDUCATION

Electrical and Electronics Engineering

2018 – 2022

Government Engineering College Thrissur

CGPA : 8.08

PROJECTS

HybridRAG-HR Semantic + BM25 Candidate Screening

Tech stack: Python, FastAPI, Postgres + pgvector (or Qdrant), BGE-M3 embeddings, BM25 hybrid search, cross-encoder reranking, Click, Streamlit

- Redesigned an HR candidate screening system around hybrid dense+sparse retrieval with cross-encoder reranking, replacing a pure vector-similarity approach and improving precision on multi-attribute queries.
- Implemented LLM-based query decomposition to translate natural-language HR prompts (e.g. "candidates who worked with Node.js and deployed on AWS") into structured metadata filters plus a residual semantic query.
- Replaced a dual-database (graph + vector) architecture with a single pgvector-backed store, cutting operational complexity while preserving relationship-style filtering via SQL.
- Built an evaluation harness (precision@k/recall@k on a golden query set) to make embedding model and reranker tuning measurable rather than ad hoc.

CERTIFICATES

- Databricks Data Engineer, 
- AWS Cloud Practitioner 